

KALEEL RAHMAN MOHAMED SATHAM

Software Engineer | Full-Stack Developer | Java Developer | AI/ML Engineer

sathamkaleel@gmail.com | +94 77 500 8136 | Kalmunai, Sri Lanka

linkedin.com/in/satham-kaleel-8ab154279 | github.com/Asterix7339 | portfolio-website-nine-ashen-78.vercel.app

PROFESSIONAL SUMMARY

Software Engineer and AI/ML Developer with project-based experience in Full-Stack Web Development, Deep Learning, Natural Language Processing, and Computer Vision. Proficient in Java, Python, and JavaScript with demonstrated expertise in React.js, Node.js, Express.js, Spring-ready MVC architecture, and MySQL database design. Built and evaluated multiple end-to-end machine learning pipelines including CNN-based medical image classification, YOLOv8 real-time object detection, and NLP-based text classification systems. Committed to Agile development practices, clean code principles, and delivering scalable software solutions. Actively pursuing a BSc (Hons) in Software Engineering while building production-quality project experience.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML5, CSS3

Frontend: React.js, React Hooks, JSP, Responsive Web Design, Tailwind CSS

Backend: Node.js, Express.js, Java EE, Servlets, REST API, RESTful Web Services, MVC Architecture

Databases: MySQL, Microsoft SQL Server, Database Design, Normalization, SQL Query Optimization

AI / ML: Machine Learning, Deep Learning, CNN, NLP, TensorFlow, Keras, Scikit-learn, YOLOv8, OpenCV, Pandas, NumPy, Matplotlib, Object Detection, Image Classification, Text Classification, Transfer Learning, Feature Engineering, Hyperparameter Tuning, Data Preprocessing, Model Evaluation

Tools & DevOps: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Eclipse, Figma, SDLC, Agile Methodology

Design: UI/UX Fundamentals, Wireframing, Prototyping, Technical Documentation

IoT: Arduino, ESP32, Sensor Integration, Real-Time Embedded Systems

PROJECTS

Harmful Religious Image Detection System | Python · YOLOv8 · Computer Vision · Deep Learning · OpenCV · Transfer Learning

- Architected a real-time object detection pipeline using YOLOv8, managing the complete ML workflow from custom dataset annotation and augmentation through model training, evaluation, and performance benchmarking across multi-class detection categories.
- Applied transfer learning on a custom-annotated image dataset, achieving measurable improvement in detection precision and significant reduction in false-positive rate compared to the baseline model through systematic hyperparameter tuning.
- Conducted structured model evaluation using standard object detection metrics (mAP, precision, recall), iterating on training configuration to optimize detection accuracy across diverse real-world image conditions.

Brain Tumor MRI Classification System | Python · CNN · TensorFlow · Keras · Deep Learning · Medical Imaging · OpenCV

- Developed a deep learning classification model using a Convolutional Neural Network (CNN) architecture — including transfer learning with pre-trained models — to classify brain MRI scans across multiple tumor categories (glioma, meningioma, pituitary tumor, and no-tumor), achieving strong validation accuracy through iterative training.
- Preprocessed and augmented a medical MRI image dataset (resizing, normalization, rotation, flipping, zoom augmentation) using TensorFlow/Keras and OpenCV to improve model generalization and reduce overfitting on the test set.
- Evaluated model performance using accuracy, precision, recall, and F1-score metrics; applied hyperparameter tuning (learning rate, batch size, dropout rate) to achieve consistent improvement across training cycles.

Hate Speech Detection System | Python · Machine Learning · NLP · Scikit-learn · TF-IDF · Pandas · NumPy

- Engineered an end-to-end NLP classification pipeline for automated hate speech detection, including text preprocessing (tokenization, stopword removal, lemmatization), TF-IDF feature extraction, and supervised ML model training using Scikit-learn.

- Trained and evaluated multiple classification algorithms (Logistic Regression, SVM, Naive Bayes) on a large-scale text dataset, selecting the best-performing model based on F1-score and precision-recall trade-off analysis.
- Implemented data cleaning and normalization workflows using Pandas and NumPy to handle noisy, real-world text data, significantly improving model robustness and evaluation scores on the held-out test set.

Ocean View Resort Reservation System | Java EE · JSP · Servlets · MySQL · REST-style API · MVC · HTML5 · CSS3 · JavaScript | github.com/Asterix7339/Ocean-View-Resort-Reservation-System

- Built a full-stack Java EE web application for hotel reservation management, implementing MVC architecture with JSP/Servlets, MySQL backend integration, and REST-style HTTP endpoint design for dynamic data exchange between frontend and server.
- Designed and implemented a normalized relational MySQL database schema supporting reservation management, room availability tracking, and admin dashboard functionality; version-controlled throughout using Git and GitHub.
- Delivered a responsive, cross-browser-compatible UI using HTML5, CSS3, and JavaScript, applying UI/UX design principles to improve usability and streamline the booking workflow for end users.

Hospital Management System | HTML5 · CSS3 · JavaScript · MySQL · Responsive Web Design

- Developed a web-based hospital management platform with modules for patient registration, appointment scheduling, doctor management, and billing; designed a normalized MySQL database schema supporting multi-entity relational operations.
- Implemented a clean, responsive front-end interface using HTML5, CSS3, and JavaScript, ensuring cross-browser compatibility, intuitive navigation, and efficient CRUD operations across all system modules.

Real-Time Fire Detection IoT System | IoT · Arduino · ESP32 · Python · Sensor Integration · Embedded Systems

- Engineered a real-time IoT fire detection system using Arduino/ESP32 microcontrollers with multi-sensor integration (temperature and smoke sensors) and an automated alert notification mechanism, achieving sub-second detection-to-alert response time.
- Integrated embedded firmware logic with sensor thresholds for reliable fire event detection; validated system performance through rigorous real-world testing scenarios.

EDUCATION

BSc (Hons) in Software Engineering | Expected: 2027

ICBT Campus, Colombo, Sri Lanka

Relevant Coursework: Software Engineering, Object-Oriented Programming, Database Systems, Machine Learning, Web Development, Data Structures & Algorithms

Higher National Diploma (HND) in Computing — Software Engineering | Completed: May 2025

BCAS Campus, Sri Lanka

Diploma in Information Technology | Completed: June 2023

BCAS Campus, Sri Lanka

EXPERIENCE

Data Entry Operator | Part-Time | Sri Lanka | 2023 – 2024

- Maintained high data accuracy and integrity across large datasets through systematic entry, validation, and quality assurance processes, consistently meeting deadlines in fast-paced environments.

PC Clinic Support Assistant | Volunteer | Sri Lanka | 2022 – 2023

- Diagnosed and resolved hardware and software issues, assisted in OS installation and peripheral configuration; developed practical IT troubleshooting and technical communication skills.

CORE COMPETENCIES

Problem Solving · Analytical Thinking · Teamwork & Collaboration · Communication · Time Management · Adaptability · Attention to Detail · Technical Documentation · Self-Learning · Agile SDLC · Version Control (Git) · Fast & Accurate Typing